Specular-Aware Monocular Depth and Clinical Adaptation Guardrails

Preliminary Findings for Colonoscopy Polyp-Size Research

Smart Wattanapornmongkol*

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Abstract

Reliable polyp-size estimation from colonoscopy video requires both useful image geometry and a defensible link from image evidence to millimetres. We report preliminary results from a self-supervised monocular depth project that uses C3VDv2 phantom data as a source-domain guardrail and REALColon clinical video as an adaptation target. The preferred source baseline is Run D, which combines SpecularNet and PPS and obtains an AbsRel of 0.1360 and an RMSE of 3.79 mm on the full C3VDv2 HF test. In contrast, unconstrained target-only adaptation produced forgetting or absolute-scale drift. Freezing depth modules preserved source outputs, but is a no-forgetting control rather than evidence of clinical depth adaptation. Teacher-anchored source replay has a promising sparse clinical-depth signal, but it has not completed target-domain, qualitative, or scale-aware selection. These findings motivate an evaluation-first, anchor-gated direct sizing endpoint; they do not demonstrate metric polyp sizing or clinical 10-mm threshold accuracy.

1 Introduction

Colonoscopy is a flexible endoscopic procedure used to inspect the large bowel. Polyps are mucosal lesions that may be removed or followed, and their estimated diameter can influence documentation, resection planning, and surveillance pathways. A useful assistive system must operate under wide-angle optics, changing working distance, specular highlights, blur, debris, bubbles, insufflation, tissue deformation, and variation in endoscope hardware.

The central technical constraint is metric scale. Under perspective projection, an image coordinate is proportional to fX/Z; scaling scene geometry and camera translation together leaves the image unchanged. Monocular depth or multi-view reconstruction can therefore yield useful relative shape without showing that a measurement is correct in millimetres. Clinical systems that make metric claims establish a scale channel explicitly, for example through lens calibration [5] or an in-scene waterjet reference [4]. This short paper reports the project state with that distinction preserved.

*Handoff contact: student ID 6738249221; phone 0639390653; LINE.
Project repository: github.com/smartwhatt/Colon-AF-SfMLearner.

2 Methods

2.1 Research setting

The implementation extends AF-SfMLearner, a self-supervised monocular endoscopy depth-and-motion framework that uses appearance flow to mitigate photometric mismatch [3]. C3VDv2 phantom data provide dense depth and pose guardrails. REALColon video provides clinical appearance diversity but not dense metric depth ground truth. Thus, C3VDv2 evaluates source-domain geometry and REALColon adaptation must be screened for both source regression and clinically meaningful scale evidence.

We compare source and adaptation branches documented in the canonical project record [1]. For every depth-adapting REALColon candidate, the required evaluation design includes C3VDv2 depth without median scaling for absolute scale, median-scaled C3VDv2 depth for relative shape, C3VDv2 pose, a fixed REALColon qualitative panel, and any available sparse clinical gate. These are selection requirements, not interchangeable endpoints.

2.2 Interpretation protocol

We use the following rules to avoid overinterpretation. A frozen module can prove that its output was preserved, but cannot establish adaptation of that module. Median-scaled depth measures relative structure and cannot be reported as metric accuracy. Sparse depth points are not lesion longest-axis labels. A metric polyp-size claim requires a leakage-checked clinical benchmark with known label provenance and a validated scale anchor or scope-specific calibration.

3 Results

Table 1 summarizes the claim-safe findings. Run D (`stage2_specular_pps/weights_best`) is the preferred C3VDv2 phantom-depth baseline. Its full HF-test metrics are AbsRel 0.1360, RMSE 3.79 mm, and $\delta < 1.25$ of 0.8318. In the tested ablation, PPS alone was weaker than the baseline, whereas the SpecularNet–PPS combination was stronger; this supports a conditional, rather than universal, benefit of PPS.

The evidence also identifies what is missing. The sparse clinical-depth signal measures a limited set of labelled points; it neither measures a lesion's longest axis nor validates 5-mm or 10-mm decision thresholds. The private candidate size archive cannot serve as headline evidence until duplicate and conflicting labels, mixed coordinate conventions, and possible split leakage are resolved.

4 Discussion

The results separate two useful but different outcomes: preserving a source geometry prior and establishing a clinical measurement endpoint. Run D should remain the phantom baseline because it is the strongest tested source-domain configuration. The target-only failures mean that REALColon adaptation should not be selected by training loss or a single target signal. Conversely, frozen depth controls are valuable because they localize forgetting, but should not be misdescribed as successful clinical adaptation.

The direct-sizing literature supports a complementary path. A single-scope, snare-anchored study has evaluated direct size regression with segmentation and depth cues [6], and MPSE reports direct per-frame regression with polyp-level aggregation [2]. These studies do not remove scale transfer as a problem, but they show that a useful endpoint need not be "reconstruct a point

Finding	Evidence	Safe interpretation
Run D is the preferred source baseline.	C3VDv2 HF test: AbsRel 0.1360, RMSE 3.79 mm, $\delta < 1.25 = 0.8318$.	SpecularNet plus PPS is the strongest tested *phantom-depth* configuration; it is not a real-tissue or polyp-size result.
PPS depends on specular handling.	PPS-only Run C is weaker than the baseline; the SpecularNet plus PPS Run D is stronger than Run B and Run A.	The tested evidence supports PPS when paired with SpecularNet, not a blanket claim for PPS.
Target-only adaptation is unsafe.	Run B and Run D target-only branches failed a planned relative-depth gate; another short full branch retained relative structure but drifted in absolute scale.	Unconstrained adaptation can damage the source prior or metric calibration.
Frozen depth is a control.	In `pose_transform` and Run B `teacher_pose_transform`, encoder/depth/specular hashes and depth outputs were unchanged.	The controls establish no forgetting for frozen modules, not clinical-domain depth adaptation.
Source replay is unselected.	A 2k replay checkpoint improved a sparse PolypSense3D raw gate (MAE 8.5612 mm) while source guardrails remained near baseline. The 3k final checkpoint is clinically interesting but awaits fixed-panel and scale-aware selection.	Replay can protect source performance during trainable-depth adaptation. It does not establish broad clinical metric depth or polyp sizing.

Table 1: Claim-safe preliminary results. All adaptation candidates remain subject to source-depth, source-pose, fixed-panel, and scale-aware gates.

cloud, then measure it." The recommended next track is a short-video direct head with lesion-level aggregation, explicit quality control, and a metric output only when a validated anchor is visible; otherwise it should return ordinal probabilities or abstain.

5 Limitations

This is an internal preliminary study, not an independent clinical validation. C3VDv2 is a controlled phantom benchmark, and REALColon lacks dense clinical depth ground truth. Reported project numbers are traceable to the canonical experiment index rather than a public checkpoint release. Sparse clinical points, private candidate labels, and fixed qualitative panels remain insufficient for metric lesion-size claims. No conclusion here establishes clinical generalization, regulatory readiness, or real-domain 10-mm threshold accuracy.

6 Conclusion

Run D is the preferred C3VDv2 phantom-depth baseline. Target-only REALColon adaptation has shown forgetting or scale drift; frozen-depth branches are no-forgetting controls; and source replay is a guarded but not yet selected clinical signal. The immediate priority is not a longer adaptation campaign. It is a target-domain selector and leakage-checked, anchor-annotated clinical evaluation that can support a direct, clip-level sizing endpoint.

References

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